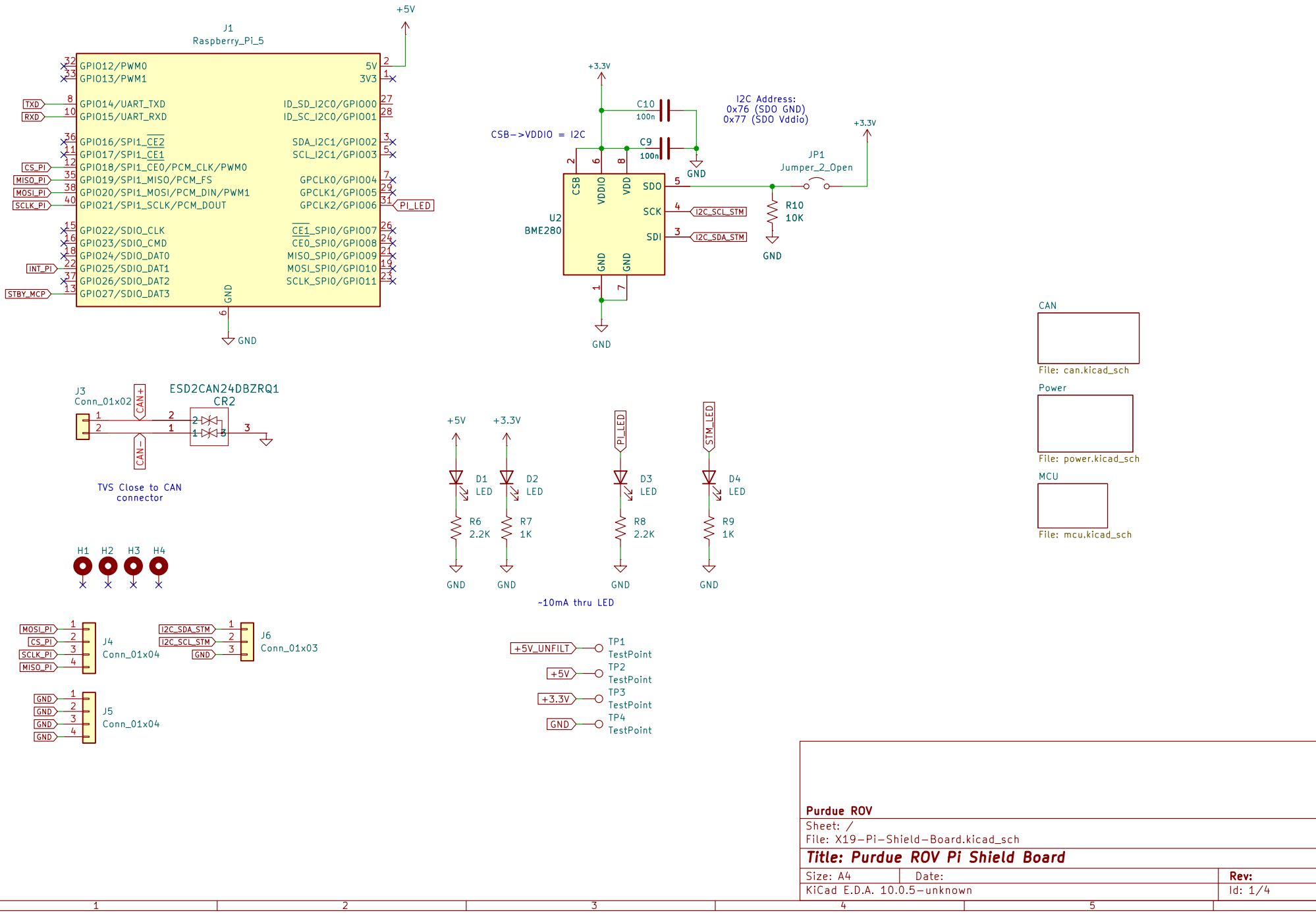
J1
Raspberry_Pi_5
+5V
GPIO12/PWM0
GPIO13/PWM1
5V
3V3
TXD
RXD
GPIO14/UART_TXD
GPIO15/UART_RXD
ID_SD_I2C0/GPIO00
ID_SC_I2C0/GPIO01
GPIO16/SPI1_CE2
GPIO17/SPI1_CE1
SDA_I2C1/GPIO02
SCL_I2C1/GPIO03
CS_PI
GPIO18/SPI1_CE0/PCM_CLK/PWM0
MISO_PI
MOSI_PI
SCLK_PI
GPIO19/SPI1_MISO/PCM_FS
GPIO20/SPI1_MOSI/PCM_DIN/PWM1
GPIO21/SPI1_SCLK/PCM_DOUT
GPCLK0/GPIO04
GPCLK1/GPIO05
GPCLK2/GPIO06
PI_LED
GPIO22/SDIO_CLK
GPIO23/SDIO_CMD
GPIO24/SDIO_DAT0
INT_PI
GPIO25/SDIO_DAT1
GPIO26/SDIO_DAT2
STBY_MCP
GPIO27/SDIO_DAT3
CE1_SPI0/GPIO07
CE0_SPI0/GPIO08
MISO_SPI0/GPIO09
MOSI_SPI0/GPIO10
SCLK_SPI0/GPIO11
GND
GND
+3.3V
C10
100n
C9
100n
GND
I2C Address:
0x76 (SDO GND)
0x77 (SDO Vddio)
CSB->VDDIO = I2C
JP1
Jumper_2_Open
+3.3V
U2
BME280
CSB
VDDIO
VDD
SDO
SCK
SDI
GND
GND
I2C_SCL_STM
I2C_SDA_STM
R10
10K
GND
J3
Conn_01x02
CAN+
CAN-
ESD2CAN24DBZRQ1
CR2
TVS Close to CAN
connector
+5V
+3.3V
PI_LED
STM_LED
D1
LED
D2
LED
D3
LED
D4
LED
R6
2.2K
R7
1K
R8
2.2K
R9
1K
GND
GND
GND
GND
~10mA thru LED
H1
H2
H3
H4
MOSI_PI
CS_PI
SCLK_PI
MISO_PI
J4
Conn_01x04
I2C_SDA_STM
I2C_SCL_STM
GND
J6
Conn_01x03
GND
GND
GND
GND
J5
Conn_01x04
+5V_UNFILT
+5V
+3.3V
GND
TP1
TestPoint
TP2
TestPoint
TP3
TestPoint
TP4
TestPoint
CAN
File: can.kicad_sch
Power
File: power.kicad_sch
MCU
File: mcu.kicad_sch
Purdue ROV
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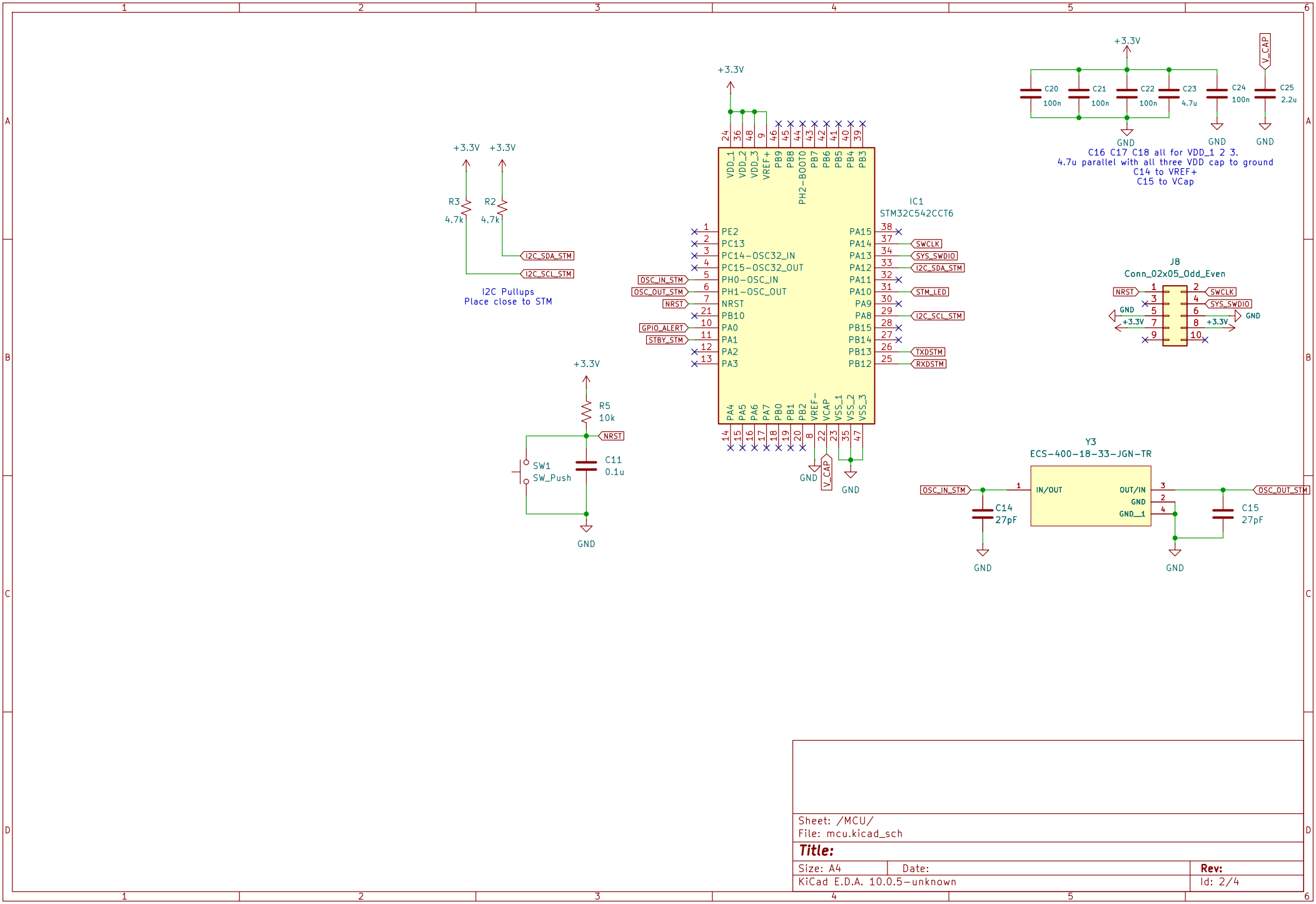
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.5-unknown
Id: 1/4


+3.3V
C20
100n
C21
100n
C22
100n
C23
4.7u
C24
100n
V_CAP
C25
2.2u
GND
C16 C17 C18 all for VDD_1 2 3.
4.7u parallel with all three VDD cap to ground
C14 to VREF+
C15 to VCap
+3.3V +3.3V
R3
4.7k
R2
4.7k
I2C_SDA_STM
I2C_SCL_STM
I2C Pullups
Place close to STM
IC1
STM32C542CCT6
VDD_1
VDD_2
VDD_3
VREF+
PB9
PB8
PH2-BOOT0
PB7
PB6
PB5
PB4
PB3
PE2
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PH0-OSC_IN
PH1-OSC_OUT
NRST
PB10
PA0
PA1
PA2
PA3
PA15
PA14
PA13
PA12
PA11
PA10
PA9
PA8
PB15
PB14
PB13
PB12
SWCLK
SYS_SWDIO
I2C_SDA_STM
STM_LED
I2C_SCL_STM
TXDSTM
RXDSTM
OSC_IN_STM
OSC_OUT_STM
NRST
GPIO_ALERT
STBY_STM
PA4
PA5
PA6
PA7
PB0
PB1
PB2
VREF-
VCAP
VSS_1
VSS_2
VSS_3
GND
V_CAP
GND
J8
Conn_02x05_Odd_Even
NRST
SWCLK
GND
SYS_SWDIO
+3.3V
GND
+3.3V
R5
10k
NRST
SW1
SW_Push
C11
0.1u
GND
Y3
ECS-400-18-33-JGN-TR
OSC_IN_STM
IN/OUT
OUT/IN
GND
GND_1
OSC_OUT_STM
C14
27pF
C15
27pF
GND
GND
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Date:
Rev:
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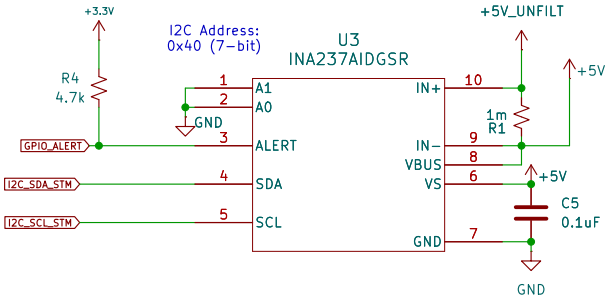

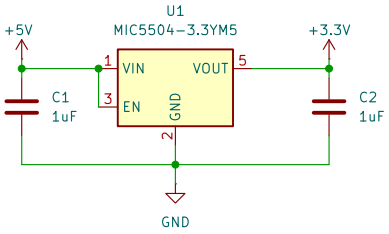

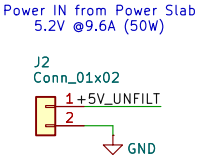

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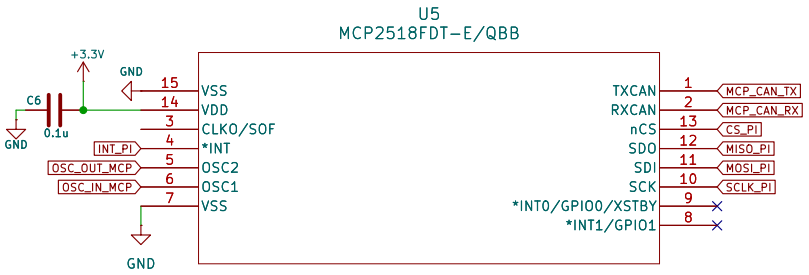

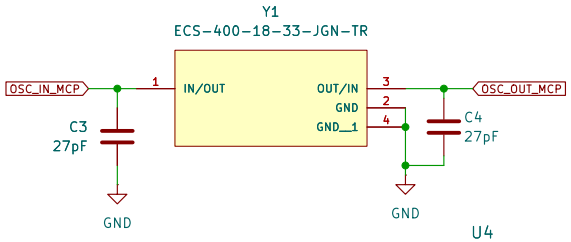

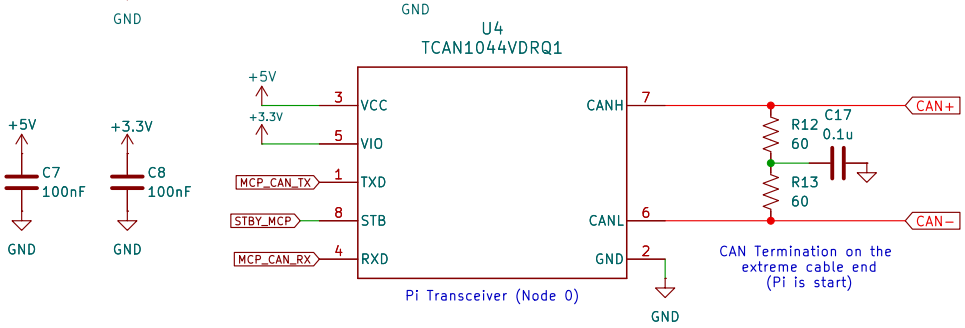

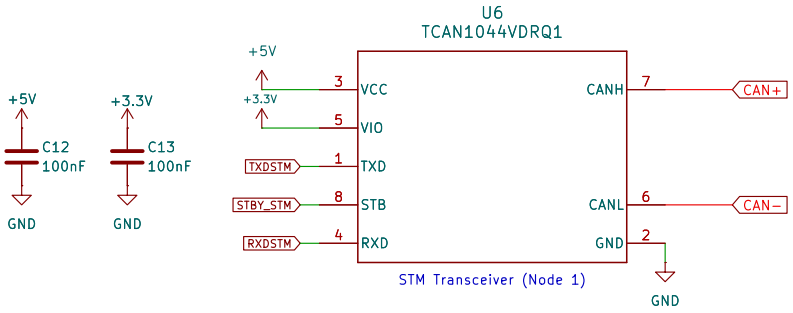

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